

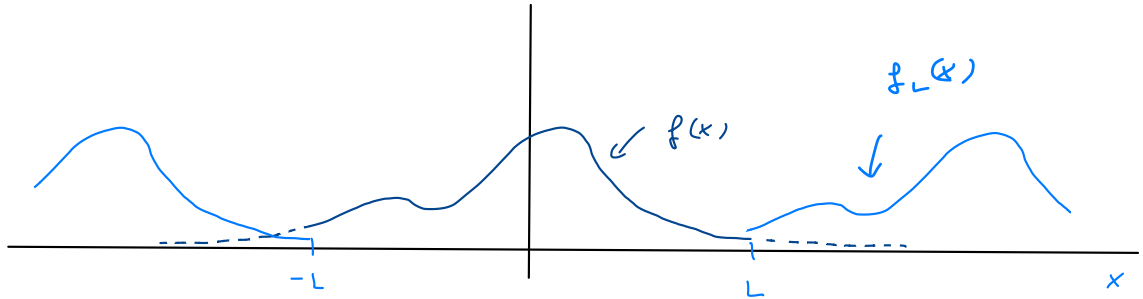
Fourier integrals

we want to extend our machinery to non-periodic functions ...

$f(x)$ nice function

① $f_L(x) \equiv f(x)$ if $|x| < L$

② $f_L(x)$ $2L$ -periodic



$$f_L(x) = a_0 + \sum_{n=1}^{\infty} \left\{ a_n \cos\left(n \cdot \frac{\pi}{L} \cdot x\right) + b_n \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \right\}$$

$$a_n = \frac{1}{L} \int_{-L}^L dy \cos\left(n \cdot \frac{\pi}{L} \cdot y\right) f(y)$$

First term: introduce $\omega_n \equiv n \frac{\pi}{L}$, $\Delta\omega \equiv \omega_{n+1} - \omega_n = \frac{\pi}{L}$

$$\Rightarrow \frac{1}{\pi} \sum_{\omega_n} \Delta\omega \cos(\omega_n x) \int_{-\infty}^{\infty} \cos(\omega_n y) f(y) dy$$

$$\xrightarrow{\frac{\pi}{L}} \int_{-\infty}^{\infty} d\omega \cos(\omega x) \frac{1}{\pi} \int_{-\infty}^{\infty} dy \cos(\omega y) f(y)$$

$$\left| f(x) = \frac{1}{\pi} \int_0^{\infty} d\omega \cos(\omega x) \int_{-\infty}^{\infty} \cos(\omega y) f(y) dy + \right.$$

$$\left. + \frac{1}{\pi} \int_0^{\infty} d\omega \dots \text{"cos" } \leftrightarrow \text{"sin"} \dots \right. \quad (*)$$

Sine and cosine transform

$$A(\omega) \equiv \frac{1}{\pi} \int_{-\infty}^{\infty} dy \cos(\omega \cdot y) f(y)$$

$$B(\omega) \equiv \frac{1}{\pi} \int_{-\infty}^{\infty} dy \sin(\omega \cdot y) f(y)$$

$$f(y) = \int_0^{\infty} d\omega \left[A(\omega) \cos(\omega \cdot y) + B(\omega) \sin(\omega \cdot y) \right]$$

example:

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$$

$$A(\omega) = \frac{1}{\pi} \int_{-1}^1 dy \cos(\omega y) \cdot 1 = \left[\frac{1}{\pi} \frac{\sin(\omega y)}{\omega} \right]_{-1}^1 = \frac{2}{\pi} \frac{\sin(\omega)}{\omega}$$

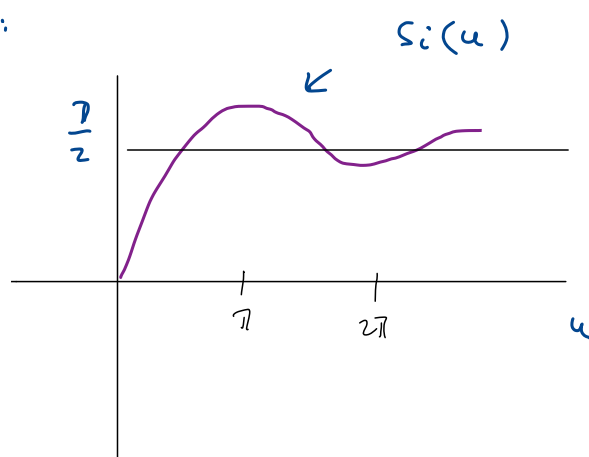
$$B(\omega) = 0$$

$$\Rightarrow f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos(\omega x) \sin(\omega)}{\omega} d\omega$$

$$x=0 \Rightarrow \int_0^{\infty} \frac{\sin(\omega)}{\omega} d\omega = \frac{\pi}{2} !$$

sine integral function:

$$\text{Si}(u) = \int_0^u \frac{\sin(x)}{x} dx$$



Insert
figure (p. 514.)

In general

$f(x)$ even $\Leftrightarrow A(\omega) \rightarrow$ cosine transform

$f(x)$ odd $\Leftrightarrow B(\omega) \rightarrow$ sine transform

Complex Fourier transform

Rewrite (*)

$$\text{use } \cos(x \cdot \omega) \cos(y \cdot \omega) + \sin(x \cdot \omega) \sin(y \cdot \omega) = \cos(\omega(x-y))$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} d\omega \int_{-\infty}^{\infty} dy \underbrace{\cos(\omega(x-y))}_{\frac{1}{2}(e^{i\omega(x-y)} + e^{-i(x-y)\omega})} f(y) =$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega e^{i\omega x} \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega y} f(y) dy}_{\hat{f}(\omega)}$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dy f(y) \cdot e^{-i\omega y}$$

Fourier
transform

$$f(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \hat{f}(\omega) e^{i\omega y}$$

inverse
Fourier

Dirac delta

$$f(x) = \int_{-\infty}^{\infty} dy f(y) \underbrace{\left(\frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega(x-y)} \right)}_{\delta(x-y)} \leftarrow \text{Dirac delta}$$

$$f(x) \equiv \int dy \delta(x-y) f(y)$$

Does $\hat{f}(\omega)$ exist?

statement: if $|f(x)|$ integrable and piecewise continuous on any finite interval $\Rightarrow \int \hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int dx e^{-i\omega \cdot x} f(x)$

notation

$$\hat{f} = \mathcal{F}(f), \quad f = \mathcal{F}^{-1}(\hat{f})$$

Example: $f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} dx e^{-x} e^{-i\omega \cdot x} = \int e^{\alpha x} \Rightarrow \frac{1}{\alpha} e^{\alpha x}$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{-1 - i\omega} e^{-x(1+i\omega)} \right]_0^{\infty} = \frac{1}{\sqrt{2\pi}(1+i\omega)}$$

A few Fourier transforms

$f(x)$	$\mathcal{F}(f)$
1	$\sqrt{2\pi} \cdot \delta(\omega)$
$e^{- x }$	$\sqrt{\frac{2}{\pi}} \frac{1}{1+\omega^2}$
$\Theta(x) e^{-x}$	$\frac{1}{\sqrt{2\pi}} \frac{1}{1+i\omega}$
$e^{-\frac{x^2}{2}}$	$e^{-\omega^2/2}$
$\chi_{[-1,1]}(x)$	$\sqrt{\frac{2}{\pi}} \frac{\sin(\omega)}{\omega}$

Properties of $\mathcal{F}$

1. $g(x) = f(-x) \Rightarrow \hat{g}(\omega) = \hat{f}(\omega)^*$
2. Fourier transformation is linear;

$$\mathcal{F}(a \cdot f + b \cdot g) = a \hat{f} + b \cdot \hat{g}$$

Proof:

$$\begin{aligned} 1. \quad \hat{g}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \, g(x) e^{-i\omega \cdot x} \\ &\quad \begin{array}{c} \uparrow \\ y = -x \end{array} \\ &\quad \begin{array}{c} \uparrow \\ f(-x) \end{array} \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dy \, f(y) e^{i\omega \cdot y} = \hat{f}(-\omega) \quad \checkmark \end{aligned}$$

2. trivial

Simple application:

$$g(x) = e^{-|x|} = f(x) + f(-x)$$
$$\uparrow$$
$$f(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

$$\begin{aligned} \hat{g}(\omega) &= \hat{f}(\omega) + \hat{f}(-\omega) = \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{1+i\omega} + \frac{1}{1-i\omega} \right) = \sqrt{\frac{2}{\pi}} \frac{1}{1+\omega^2} \end{aligned}$$

$$\mathcal{F}\{e^{-|x|}\} = \sqrt{\frac{2}{\pi}} \frac{1}{1+\omega^2}$$

$\nearrow$ exponential $\uparrow$ Lorentzian

Def.: convolution

$$h \equiv f * g$$

$$h(x) = \int dy \ f(\underbrace{x-y}_{\substack{\uparrow \\ \text{sum} = x!}}) \underbrace{g(y)}_{\substack{\uparrow \\ \text{sum} = x!}} = \int dy \ f(y) g(x-y)$$

$$f * g = g * f$$

$$(f * g) * h = f * (g * h) = \dots \equiv f * g * h$$

6. f, g piecewise continuous, bounded
absolutely integrable

$$\mathcal{F}[f * g] = \sqrt{2\pi} \hat{f}(\omega) \cdot \hat{g}(\omega)$$

~ F.T. of convolution $\rightarrow$ product of
FT's

Proof:

$$\sqrt{2\pi} \cdot \mathcal{F}[f * g](\omega) = \int_{-\infty}^{\infty} dx \ e^{-i x \cdot \omega} \int dy \ f(y) \cdot \underbrace{g(x-y)}_{\substack{= \\ y}}$$

$$= \int_{-\infty}^{\infty} d\gamma \int_{-\infty}^{\infty} dy \ e^{-i(\gamma+y) \cdot \omega} f(y) \cdot g(\gamma) =$$
$$e^{-i\gamma \cdot \omega} \cdot e^{-iy \cdot \omega}$$

$$= \underbrace{\left(\int_{-\infty}^{\infty} d\gamma \ e^{-i\gamma \cdot \omega} g(\gamma) \right)}_{\sqrt{2\pi} \hat{g}(\omega)} \cdot \underbrace{\left(\int_{-\infty}^{\infty} dy \ e^{-iy \cdot \omega} f(y) \right)}_{\sqrt{2\pi} \hat{f}(\omega)}$$

□

Discrete FT

$f(x)$ 2π -periodic

discretize $f_k \equiv f\left(x = \frac{k \cdot 2\pi}{N}\right) \quad k = 1, \dots, N-1$

$$f(x) = \sum_n c_n e^{i n \cdot x}$$

• truncate : $n = 0, \dots, N-1$

• interpolate at x_k, f_k

$$f_k = \sum_{n=0}^{N-1} c_n \underbrace{e^{i n x_k}}_{e^{i \frac{n \cdot k}{N} \cdot 2\pi}} = \bar{w}^{n \cdot k}$$

claim

$$c_m = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-i m \cdot x_k}$$

$$\begin{aligned} \hat{f}_n &\equiv \sum_{k=0}^{N-1} f_k e^{-i \frac{n \cdot k}{N} 2\pi} \\ f_k &= \frac{1}{N} \sum_{n=0}^{N-1} \hat{f}_n e^{i \frac{n \cdot k}{N} 2\pi} \end{aligned}$$

Remarks:

$$\cdot \quad F_{nk} \equiv e^{-i \frac{2\pi}{N} \cdot n \cdot k} = w^{n \cdot k} \quad (w = e^{-\frac{2\pi}{N} \cdot i})$$

$$\hat{\underline{f}} = \underline{\hat{F}} \cdot \underline{f}$$

$$\underline{f} = \frac{1}{N} \underline{\hat{F}}^* \cdot \hat{\underline{f}} = \underbrace{\left(\frac{1}{N} \underline{\hat{F}}^* \cdot \underline{\hat{F}} \right)}_{\underline{I}_N} \underline{f} \quad \frac{1}{N} \underline{\hat{F}} \text{ unitary}$$

$$\bullet \quad N = 2^M \rightarrow \exists \text{ FFT} \quad \sim \text{run time} \sim M \ln M$$